POLICY BRIEF · TEACHER LABOR MARKETS · JULY 2026

Who leaves teaching? Two decades of teacher attrition in the United States, 2005–2025

Evidence from 174,872 school teachers followed for twelve months each in linked Current Population Survey microdata

EXECUTIVE SUMMARY

- Linking 8.2 million individuals across twenty annual CPS rotation panels, **17.6% of degree-holding school teachers are no longer teaching twelve months later**: 10.8 percentage points (pp) move to another occupation, 5.6 pp leave the labor force and 1.2 pp are unemployed.
- Attrition has **risen by about five points in two decades** — from 15–16% in 2005–2010 to 19–20% in 2022–2024, with a COVID spike of 21.8% in 2019→2020.
- **Not all exits are permanent: one in six leavers (15.7%) is back teaching within three months** — part of measured attrition is short-run churn, slightly more common among women (16.1%) than men (14.0%).
- **Working part-time is by far the strongest predictor** of leaving ($+14.5$ pp). Exit risk is U-shaped in age (lowest at 40) and higher for Black ($+6.5$ pp), non-citizen ($+8.0$ pp) and preschool/kindergarten ($+4.0$ pp) teachers. Once the sample is restricted to degree holders, a graduate degree adds only a modest extra protection (-1.3 pp).
- The **typical leaver** is a woman in her late career, working part-time, without a graduate credential — the top predicted-risk decile faces a 40% exit probability, 2.3 times the average teacher.

17.6 %
of teachers leave the profession within 12 months

$+$14.5 pp
extra exit risk from part-time work

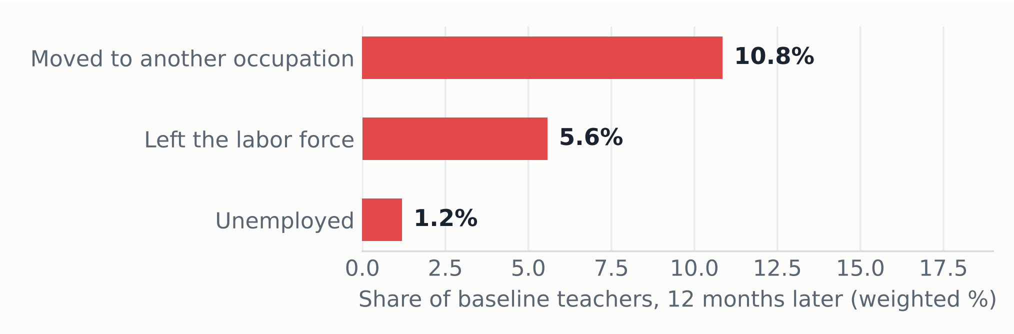
1 in 6
leavers is back teaching within 3 months

1. Data: twenty annual panels of teachers

We use official **Current Population Survey (CPS) basic monthly microdata**: 251 monthly files covering January 2005 to December 2025, from the U.S. Census Bureau (2024–2025) and the NBER mirror (2005–2023); October 2025 was never released. Exploiting the CPS 4–8–4 rotation design, each adult interviewed in year t is re-interviewed in the same calendar month of $t{+}1$ and can be linked by household and person identifiers, with matches validated on sex, race and age progression (Madrian–Lefgren criterion). This yields **8.2 million validated individual links**, of which **174,872 were school teachers** at baseline — preschool to secondary, including special education (occupation codes 2300–2330, identical across the 2002, 2010 and 2020 classifications) — **holding at least a bachelor's degree**, the standard restriction in the attrition literature. A *leaver* is a baseline teacher who, twelve months later, is not employed as a school teacher. Rates use CPS final person weights.

Caveat: occupation in the returning rotation is re-coded independently, which inflates measured occupation switching; levels should be read as upper bounds, while *comparisons across groups and over time* — the object of this brief — are far less affected.

Figure 1: Most leavers move to other jobs rather than out of work. Destination twelve months later, share of baseline teachers (weighted), 2005–2025.



Source: authors' calculations, linked CPS basic monthly files 2005–2025.

2. Attrition is rising — and part of it is churn

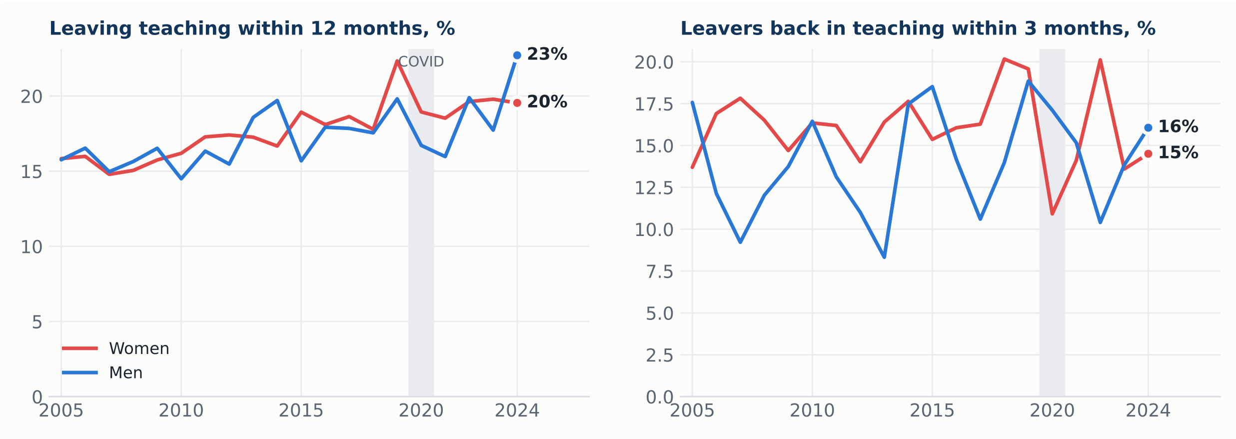
Attrition climbed from 15–16% in 2005–2010 to 19–20% in 2022–2024 (Figure 2, left). The 2019→2020 pandemic year stands out at 21.8%. Gender gaps in leaving are small and unstable; in the most recent year men (22.7%) actually exceeded women (19.5%).

The rotation design also lets us re-observe leavers for up to three months after we see them out of teaching: **15.7% are already teaching again** (women 16.1%, men 14.0%; Figure 2, right). A meaningful slice of measured attrition is therefore short-run churn — temporary exits, between-school moves timed around the survey, or misclassification — rather than permanent loss.

3. Who leaves: raw differences

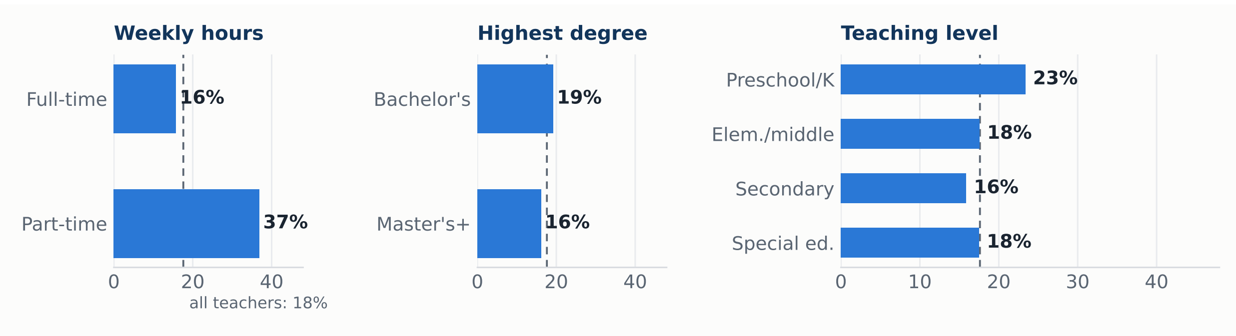
Attrition concentrates in three margins: **hours**, **teaching level** and, more weakly once degree holders only are considered, **credentials**. Part-time teachers leave at more than twice the rate of full-time teachers; preschool/kindergarten is the most exposed level.

Figure 2: Attrition has risen five points in two decades; one in six leavers returns within three months. Weighted rates by base year and gender.



Source: authors' calculations, linked CPS 2005–2025. Return window truncated by the CPS design at three monthly interviews after $t{+}12$ (leavers re-interviewed at least once: 21,873 of 30,244).

Figure 3: Attrition more than doubles without full-time hours. Weighted 12-month attrition rate by baseline characteristics, 2005–2025.



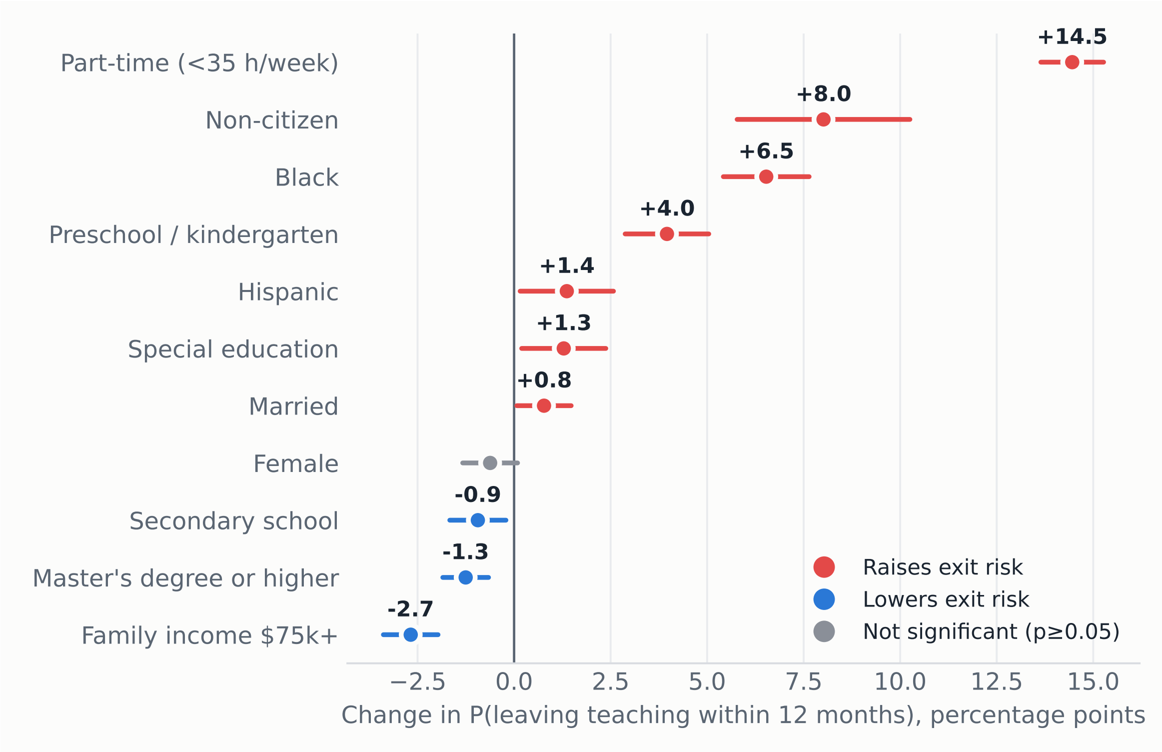
Source: authors' calculations, linked CPS 2005–2025. Dashed line: all-teacher average (17.6%).

4. A probit model of leaving teaching

We estimate a probit for the probability of leaving, with covariates measured at baseline, base-year fixed effects, and standard errors clustered by household (Table 1); Figure 4 plots the average marginal effects. Three results stand out. First, **part-time status raises exit probability by 14.5 pp**, the largest effect in the model by an order of magnitude — attachment to full-time hours, not demographics, is the dominant margin. Second, **minority and non-citizen teachers are significantly more likely to leave** (Black $+6.5$ pp, non-citizen $+8.0$ pp, Hispanic $+1.4$ pp), a retention gap with direct implications for workforce diversity. Third, among degree holders **a graduate degree adds only -1.3 pp of protection**, while higher family income (-2.7 pp) and teaching secondary school (-0.9 pp) also retain; gender is not significant once hours and demographics are held fixed.

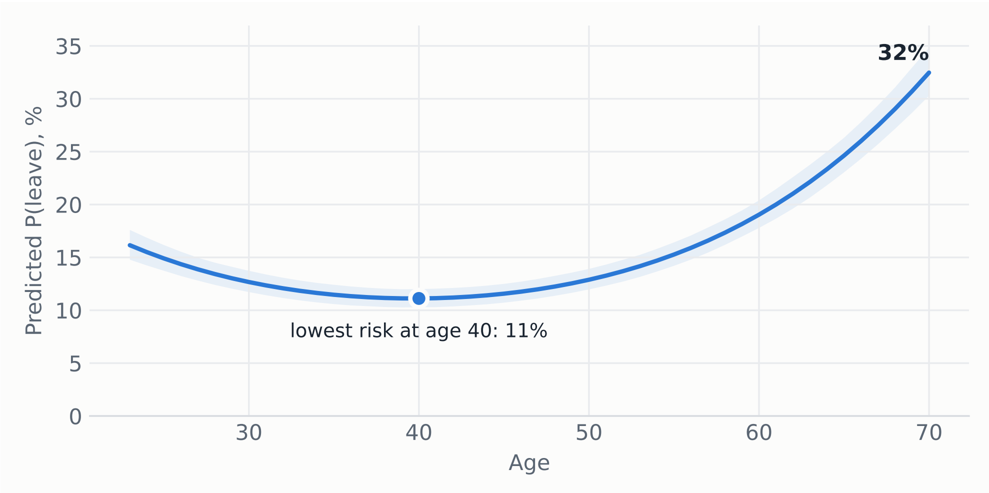
Age enters with a robust U-shape (Figure 5): predicted risk falls to a minimum of 11% around age 40 and accelerates after 55 as retirement margins open, reaching 32% at 70 — the familiar U-shaped attrition profile of the teacher-turnover literature.

Figure 4: Hours dominate demographics. Average marginal effects on the probability of leaving teaching within 12 months, with 95% confidence intervals.



Source: probit of Table 1. Age terms shown in Figure 5.

Figure 5: Exit risk is U-shaped in age, lowest near 40. Predicted probability of leaving by age, other covariates at sample means; shaded band: 95% CI.



Source: probit of Table 1, simulation-based confidence band.

5. The profile of the typical leaver

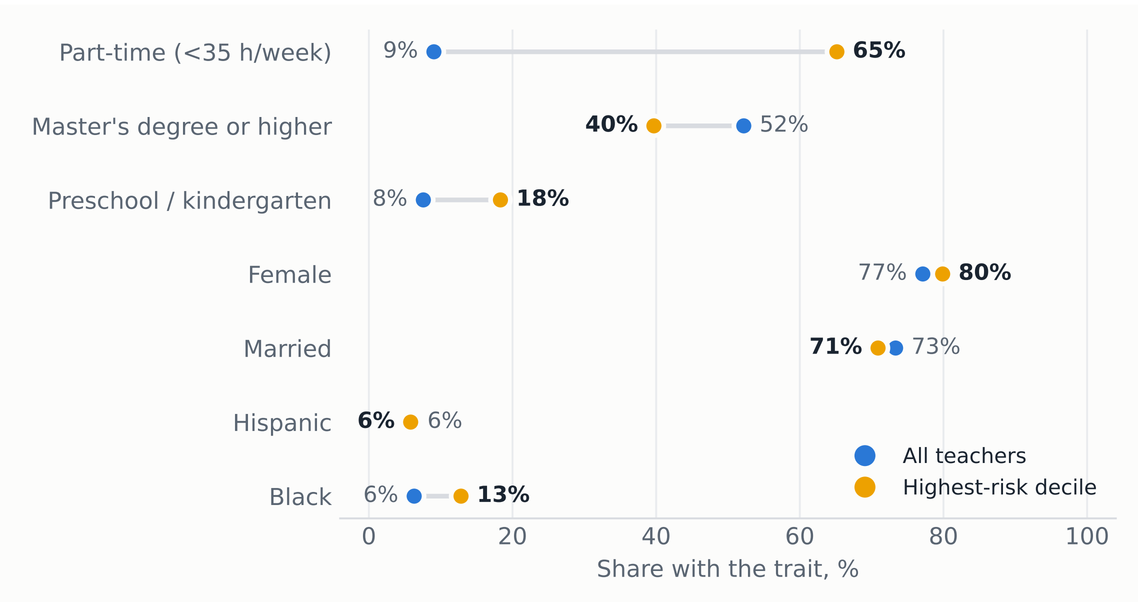
Ranking teachers by predicted risk, the top decile — with a mean exit probability of 40%, against 17% for the average teacher — is overwhelmingly female (80%), in the late-career age range (mean 55), seven times more likely to work part-time (65% vs. 9%), less likely to hold a graduate degree (40% vs. 52%), and over twice as concentrated in preschool/kindergarten (18% vs. 8%), with Black teachers over-represented (13% vs. 6%).

Table 1: Probit estimates: probability of leaving teaching within 12 months, 2005–2025.

	Probit coef.	(SE)	AME (pp)
Age (years)	-0.066***	(0.004)	-1.59***
Age^2/100	0.083***	(0.004)	+2.01***
Female	-0.026*	(0.015)	-0.62*
Married	0.032**	(0.015)	+0.77**
Black	0.271***	(0.024)	+6.53***
Hispanic	0.057**	(0.026)	+1.37**
Non-citizen	0.333***	(0.047)	+8.01***
Master's degree or higher	-0.052***	(0.013)	-1.25***
Part-time (<35 h/week)	0.601***	(0.018)	+14.46***
Hours not reported	0.378***	(0.019)	+9.10***
Family income $75k+	-0.111***	(0.015)	-2.68***
Preschool / kindergarten	0.165***	(0.023)	+3.96***
Secondary school	-0.039**	(0.015)	-0.94**
Special education	0.053**	(0.023)	+1.29**
Constant	0.121	(0.081)	
Base-year fixed effects		Yes	
Observations		174,872	
Pseudo R^2		0.056	

Cluster-robust standard errors (by household) in parentheses. AME: average marginal effect in percentage points. Sample: school teachers with a bachelor's degree or higher. Reference categories: full-time, bachelor's degree, elementary/middle school. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 6: The highest-risk teacher: part-time, less credentialed, early-years. Characteristics of the top predicted-risk decile vs. all teachers.



Source: predicted probabilities from the probit of Table 1.

6. Policy implications

The results locate teacher attrition in **weak attachment** — part-time hours, the early-childhood segment, both career extremes — rather than in broad demographics, and show a **structurally rising trend** since the early 2010s. Four levers follow. (i) **Full-time conversion** of involuntary part-time positions attacks the single largest risk factor. (ii) **Retention packages for preschool and kindergarten teachers**, the most exposed level. (iii) The significant **Black, Hispanic and**

non-citizen retention gaps warrant targeted monitoring, since they compound into the diversity of the future workforce. (iv) With **one in six leavers back within three months**, cheap re-entry pathways — fast-track rehiring, credential portability across districts, return bonuses — can recover part of the outflow at low cost.

Methods in one line. Twenty linked CPS 4–8–4 rotation panels, 2005→2025 · 174,872 teachers · probit, year FE, household-clustered SE · fully reproducible: `src/02_build_cps_panel.py` → `src/05_evolution.py`.